

A 1 MHz Buck Converter with Type-III Compensation

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Abstract – 1 MHz voltage-mode controlled buck converter. By using a differential difference amplifier (DDA) and OTA, a novel Type-III compensator is implemented to guarantee stability. Moreover, based on the unique structure of the proposed compensator, an end-point prediction (EPP) scheme is also implemented to achieve fast reference-tracking responses. The circuit utilizes a 1 MHz reference clock and has a peak efficiency of 92%. The measured maximum output power is 3 W with 1 V output voltage and 3 A load current.

I. INTRODUCTION

For our PMIC final project, we decided to implement a high efficiency, high load buck converter. The primary specification was to achieve 90% across our load range of 2-3 A. For the buck converter to function as intended, we aimed for 1% ripple voltage. A type III compensation network was implemented to ensure stability across the full load range. The full architecture of the system is shown below in Figure 1.

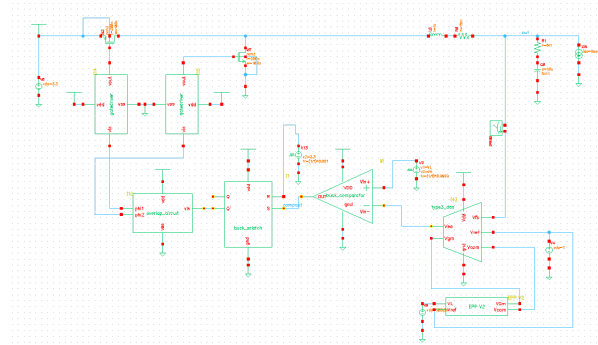


Figure 1. Full schematic for compensated feedback buck converter.

The work was implemented using the thick oxide 3.3V 180nm PDK that was provided to ensure correct operation with a high supply voltage.

II. DESIGN OF DDA-BASED TYPE-III COMPENSATOR

A. Review of Type-III Compensators

A compensator network serves two purposes for the buck converter: 1.) detect errors in the output voltage with negative feedback, and 2.) introduce low-frequency zeros to stabilize that feedback.

The RLC filter in the main buck circuit has a resonant frequency at

$f_0 = \frac{1}{2\pi\sqrt{LC}} \approx 50.3 \text{ kHz}$. For proper stability compensation, zeros must be introduced at frequencies lower than f_0 , with poles at higher frequencies. Specifically, a zero f_z reduces the phase response by 90° for $f > f_z$, while a pole f_p increases the phase by 90° for $f > f_p$; the Barkhausen stability criterion requires that the phase does not reach 180° (i.e. positive feedback) with a gain greater than 0 dB (unity gain).

B. Design of DDA-Based Type-III Compensator

For this project, a Type-III compensation scheme was implemented using an operational transconductance amplifier (OTA) followed by a differential-difference amplifier (DDA), as shown in Figure 2.

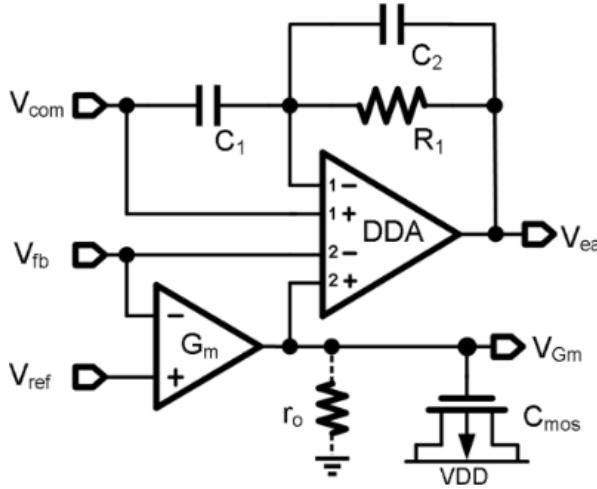


Figure 2. Schematic for OTA/DDA-based Type-III compensator [1].

The OTA, with transconductance gain G_m , converts the voltage error into an output current, which is then fed into an output resistor r_o . This creates a zero-pole pair according to

$$z_1 = \frac{G_m}{C_{mos}}; p_1 = \frac{1}{r_o C_{mos}} \quad (\text{Equation 1})$$

In order to provide adequate open-loop voltage gain $H_0 = G_m r_o > 1$, placing p_1 at a low frequency is a necessary tradeoff. Then the DDA supplies additional zero-pole compensation via second-order RC feedback:

$$z_2 = \frac{1}{(C_1 + C_2)R_1}; p_2 = \frac{1}{C_2 R_1} \quad (\text{Equation 2})$$

The theoretical zero and pole locations are listed in Table 1, along with their associated component values.

OTA: $C_{mos} = 16 \text{ pF}$	
$G_m = 3.29 \text{ } \mu\text{S}$	$z_1 \approx 32.7 \text{ kHz}$
$r_o = 500 \text{ k}\Omega$	$p_1 \approx 19.9 \text{ kHz}$
DDA: $R_1 = 3.2 \text{ k}\Omega$	
$C_2 = 2.25 \text{ pF}$	$p_2 = 22.1 \text{ MHz}$
$C_1 = 1 \text{ nF}$	$z_2 = 49.6 \text{ kHz}$

Table 1. List of passive component values in the compensator, and zero/pole values controlled by those passive components.

C. Design of Differential-Difference Amplifier (DDA)

For our DDA design we used a pmos input stage with class AB Transconductor stages. These provide the gain for our system to drive the comparator input and drive the compensation network. These stages then feed into low voltage current mirrors to maintain wide operation throughout the rails. We scaled our output stage such that we can drive higher loads. This didn't affect the speed of the DDA as it had a flat loop gain and only dropped off around 10 MHz. The schematic also includes a bias generator for the cascades in this design. The bias keeps the transistors in saturation and tracks the current fed into the differential bias current.

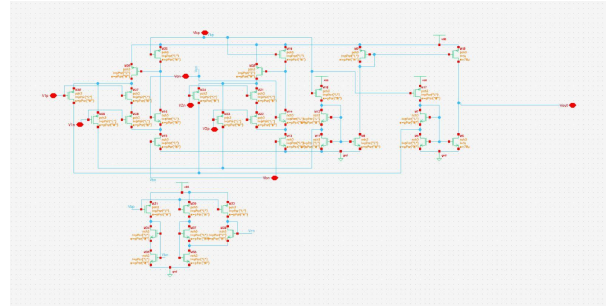


Figure 3. Schematic for DDA.

We tested the functionality of the DDA with various operational tests. The balun test ensures the amplifier correctly amplifies differential signals while rejecting common-mode inputs, validating gain, symmetry, and CMRR. Below we can see the balun test follows expected results, but we do have a limit.

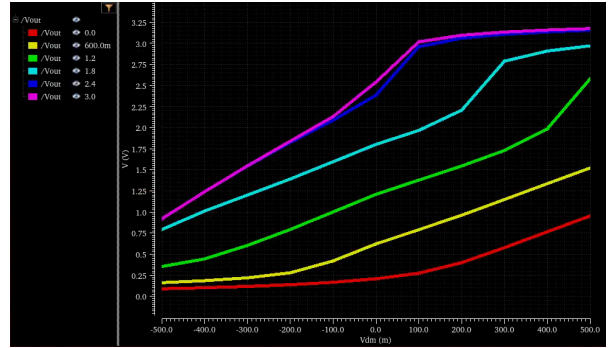


Figure 4. Balun test results.

The Common Mode Detector tests the same output signal feeding back to the DDA differential inputs. This tests through the full rail, so we can see the exact point at which the cascode breaks with each reference. Again, this gives us good results with the linearity through the lower inputs.

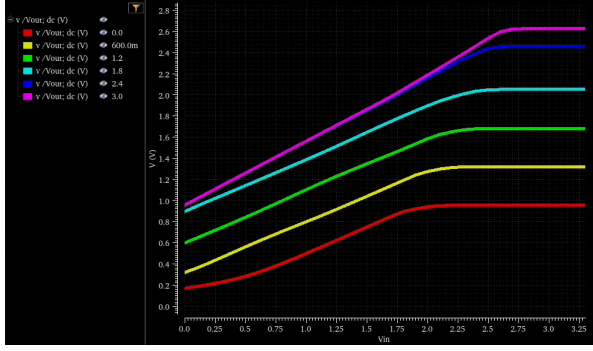


Figure 5. Common mode detection results.

We also conducted tests for the averager, doubler, and inverter configurations, which all resulted in a large linear region for the DDA to operate in. We also checked the IV characteristics of the DDA which operate well with the current a little higher than expected.

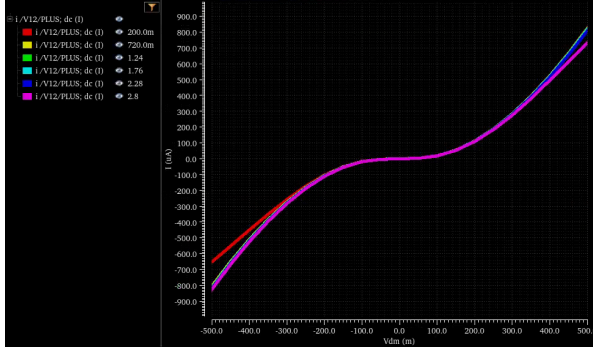


Figure 6. DDA IV characteristic. The current is higher due to the higher drive strength we implemented in the DDA.

D. Design of Transconductance Amplifier (OTA)

The OTA, as its name suggests, outputs a current proportional to the input voltage difference. Each of the two gate inputs has a CMOS pair to convert voltage to current; addition and subtraction of these currents is performed by current mirrors. This OTA topology has an advantage over other op-amp structures: the push-pull current paths allow for full rail-to-rail voltage headroom (instead of turning off when V_{TH} thresholds are not met).

In the context of the compensator, the OTA serves two purposes: 1.) to compare the output voltage against a fixed reference, and 2.) to provide a low-frequency zero and cancel out the poles in the buck RLC filter (in accordance with Eqs. 1-2). The reference voltage is chosen to be $V_{ref} = 1$ V, for direct comparison with the 1 V output.

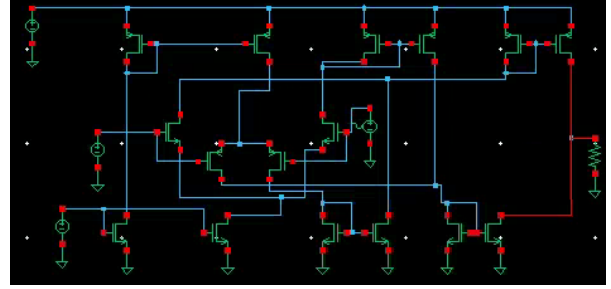
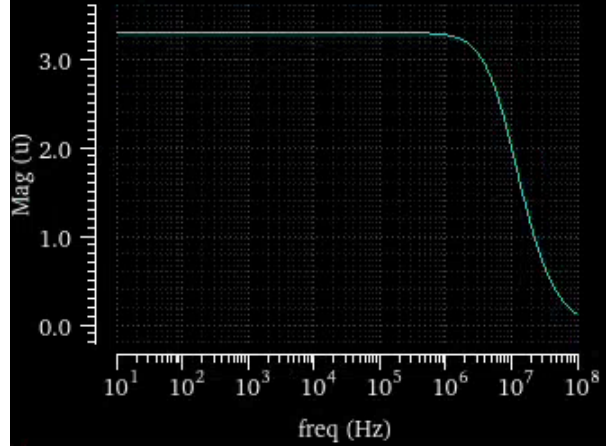
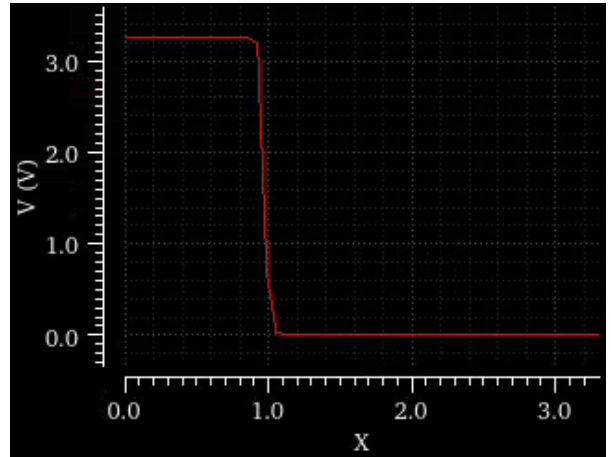


Figure 7. Full schematic for OTA.

Figure 8. Frequency response of transconductance; the results indicate that $G_m = 3.29 \mu\text{S}$ at a 1 dB bandwidth of 28 MHz.Figure 9. DC response of V_{Gm} (y-axis) with respect to V_{fb} (x-axis) for $V_{ref} = 1$ V; at equilibrium, V_{Gm} settles to 1 V.

III. END-POINT PREDICTION (EPP) SCHEME

A. Justification for an EPP and Derivations

In ideal operating conditions, the reference voltage V_{ref} and the ramp waveform maximum (V_H) and minimum (V_L) are both constant. However, their values cannot be assumed to be constant in a real-world application, where both the reference and the ramp waveform would be generated using non-ideal circuits with offset voltages, temperature sensitivities, and other sources of error. Given that a number of quantities related to the DDA and OTA have mathematical relationships with one another, there is a valid concern that small fluctuations can disrupt the high-level control scheme. The EPP exists to both define V_{com} according to an equation relating it to the ramp waveform and V_{ref} (for a buck converter operating in CCM), and to help keep the values of V_{Gm} and V_{ref} relatively tied together using a fluctuation-sensitive auto-correction circuit.

To further motivate the need for the EPP, it should be explained what V_{com} is. V_{com} is connected to the V_{1+} input of the DDA, and V_{1-} is connected to V_{ea} (the output of the error amplifier) with a feedback path. In steady state, the values of V_{1+} and V_{1-} are (ideally) equal, and this steady-state value is primarily defined by the set value of V_{1+} . As such, V_{com} defines the steady-state output of the error amplifier V_{ea} . To therefore define and obtain V_{com} , an equation must be derived according to Man Siu et al. [2].

Since V_{ref} equals the output of the resistive potential divider connected to the output voltage V_{out} , the following is true:

$$V_O = \frac{V_{ref}}{B} \quad (\text{Equation 3})$$

Where B is the resistor ratio of the potential divider (in other words, the conversion ratio). The duty cycle D is also related to the output of the error amplifier and the maximum/minimum voltages of the ramp waveform as follows:

$$\frac{V_O}{V_I} = D = \frac{V_{ea} - V_L}{V_H - V_L} \quad (\text{Equation 4})$$

Knowing that $V_{ea} = V_{com}$ in the steady-state, the above two equations can be combined to yield the following:

$$V_{com} = \left(\frac{V_H - V_L}{B V_I} \right) V_{ref} + V_L \quad (\text{Equation 5})$$

Since $V_H = 3.3 \text{ V}$, $V_L = 0 \text{ V}$, $B = 1$, and $V_I = 3.3 \text{ V}$ are all

set by the design, this equation can therefore be implemented in design through use of two additional OTAs with gains equal to the coefficients of V_{ref} and V_L (just 1). Their currents can be summed and sunk by an appropriately sized resistor to produce the required V_{com} that changes accordingly even if V_{ref} and the ramp waveform (V_H and V_L) fluctuate.

Secondly, the steady-state output voltage of the compensator OTA, V_{Gm} , follows the value of V_{ref} . However, stray capacitance can cause this response to be very slow. As such, the EPP also implements a circuit that detects any significant fluctuations in the reference voltage and quickly shorts V_{ref} to V_{Gm} in order to bypass this normally slow response.

The high-level schematic of the EPP is shown in Figure 10.

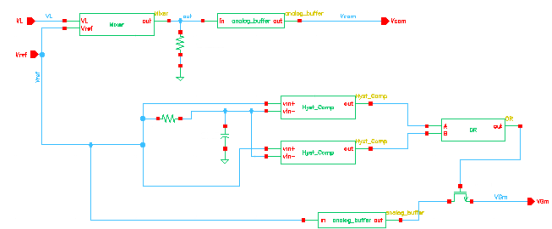


Figure 10. EPP schematic, with V_{com} current mixing (top) and V_{Gm} - V_{ref} correction (bottom).

B. Design of the EPP Circuitry

The current mixing OTAs for the creation of V_{com} are shown below in Figure 11.

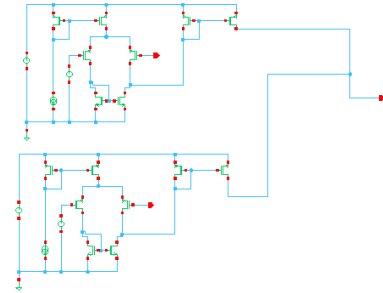


Figure 11. Detail of current mixing circuitry.

The concept is to set each OTA's gain to the value of the coefficient of its term in the V_{com} equation. These currents each enter the OTA's individual output current mirror, whose opposite outputs are connected together

to perform current summation. Note the use of PMOS differential pairs, which was done due to having relatively low input voltages that would make saturating an OTA with an NMOS differential pair difficult. The current mirrors were also chosen since their output nodes have a high resistance that allows the current mirrors to act like nearly ideal current sources and sum their currents without interfering with one another. A notable trade-off was in choosing the value of the mixing resistor that produces a voltage proportional to the sum of the currents. If the resistor was too large, then any stray capacitance could delay the natural adjustment of V_{com} to a significant degree. If the resistor was too small, then the OTAs would have to instead provide larger currents, overall increasing the power dissipated.

The V_{ref} - V_{Gm} tracking circuit operates by continually passing the value of V_{ref} into one input of each of two hysteretic comparators (but opposite inputs). Simultaneously, V_{ref} is passed into an RC delay network, whose output is passed into the other input of each comparator. As a result, one comparator monitors whether the delayed V_{ref} is significantly larger than the non-delayed V_{ref} , while the other comparator monitors whether the delayed V_{ref} is significantly smaller than the non-delayed V_{ref} . If either is true, an OR gate produces a logic high output that turns on a pass transistor to connect the changing V_{ref} to V_{Gm} .

An obvious trade-off is the amount of hysteresis in the comparators (too little produces constantly-on comparators, too much cannot detect significant fluctuations). Another is the size of the pass transistor (too large produces a large gate capacitance whose delay degrades response time, and too small has a large resistance that can produce a systematic voltage offset). A third and more subtle trade-off was the sizing of the RC delay network. It needs to provide enough delay to allow a fluctuation in V_{ref} to be properly captured, but not too much delay such that the response occurs well after the fluctuation.

The performance of the V_{ref} - V_{Gm} tracking can be seen in Figure 12.

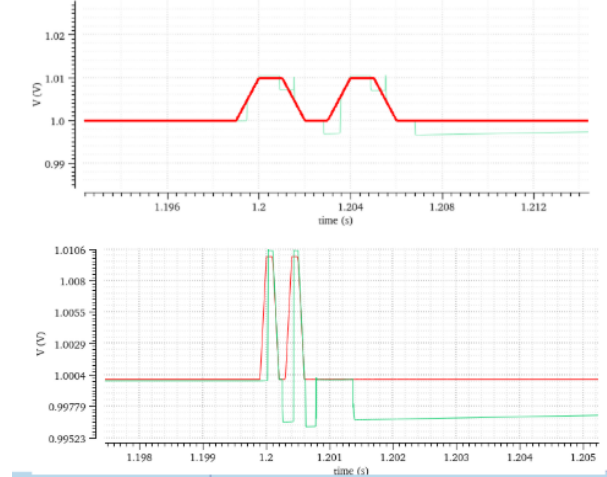


Figure 12. Results of EPP error tracking.

One case shows spikes in V_{ref} on a 1 ms time scale (top), and the other shows spikes on a 100 μ s time scale (bottom). In both cases, the output of the pass transistor (green) follows V_{ref} (red) when the two reach a difference of more than 10 mV. This avoids having an oversensitive response. However, the circuit makes a sacrifice in that it cannot detect and correct extremely short fluctuations (say, on 1 ns) due to undershoot error. This error increases as the time scale becomes smaller and takes a long time to settle.

The dissipated power of the EPP is 9.51 mW.

IV. COMPARATOR

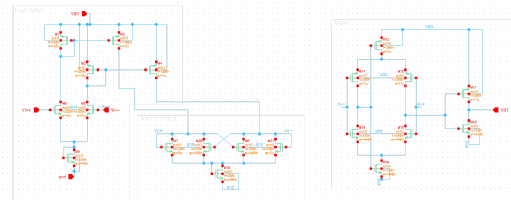


Figure 13. Comparator schematic.

The comparator block is designed to generate the PWM signal based on the input of a ramp wave and the output of the error amplifier, or in this case, the DDA. The block was optimized for a combination of speed, power consumption, and valid input swing to ensure functionality for this application. A 3-stage comparator design was implemented for this project. The first stage is a preamplifier stage, and features a typical differential pair with tail current source, mirroring the

signal over to the next stage. The latch based decision circuit stage ensures that only one output will be high. Finally, the last stage buffers the output, increases device width for drive strength, and also includes a decision element.

As shown in Figure 14, the comparator has a delay of about 5ns during testing, and a power consumption of $238\mu\text{W}$.

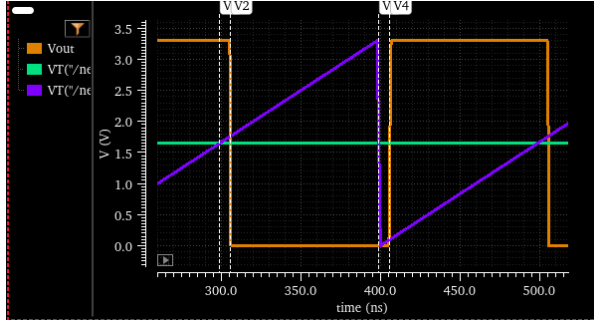


Figure 14. Transient results for comparator.

V. SR LATCH

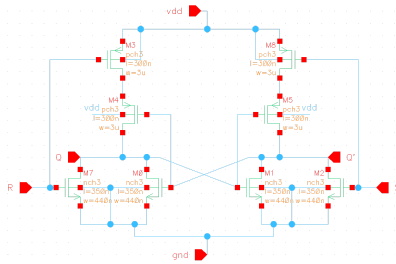


Figure 15. Full schematic for SR latch.

The SR (set-reset) latch is implemented to promote cyclic, event driven switching and guarantee the PWM waveform is consistent. A NOR based SR latch was implemented for this project. Similar to the comparator, the block was optimized for speed and power consumption, resulting in a $1.29\mu\text{W}$ power consumption and 0.15ns delay.

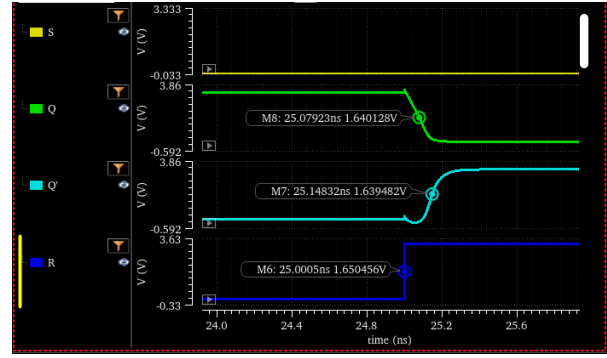


Figure 16. SR latch performance.

VI. DEADTIME CONTROL

As efficiency is prioritized in the buck converter design, a digital circuit was controlled to prevent shoot-through losses in the PFETs. At high switching speeds this can quickly degrade the efficiency and damage the converter. In order to prevent this, a non-overlapping circuit with capacitive delay was utilized to set a controlled time between turning on the PFETs.

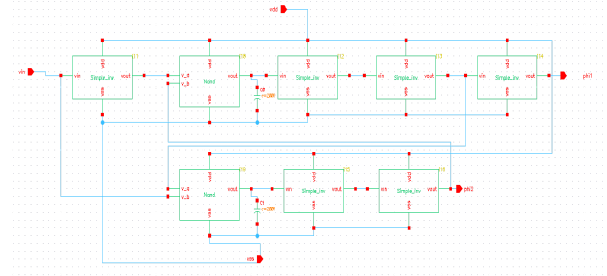


Figure 17. Non-overlapping circuit.

The proposed circuit introduces a capacitor as well as gate inverters to introduce a delay in the PFETs. In order to optimize the Dead-time, a sweep on the value of the capacitor was performed, resulting in a value of 1.75pF. This result yielded a dead-time of approximately 2.5ns, or a total dead-time of 0.5% with a 500ns switching period.

The overall propagation delay of the dead-time control subcircuit was less than 3ns, well within the switching period of the system.

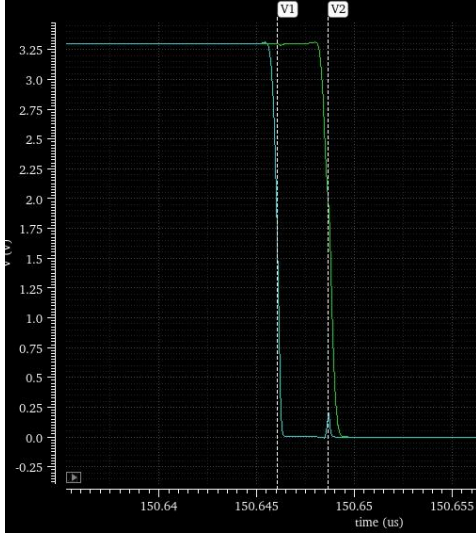


Figure 18. Deadtime performance.

VII. GATE DRIVER

A well-designed gate driver must provide enough capacitive drive strength to charge and discharge the PFETs, as well as minimize switching/driving losses.

$$P_{sw} = \frac{1}{2} V_{ds} I_d (t_r + t_f) f_{sw}$$

$$P_{gate} = (1 + \frac{1}{k}) C V^2 f_{sw}$$

To prioritize efficiency, a large step-up factor of 25 was used. This choice allowed for smaller gate driving losses as well as being within the margin with a 500ns period. The subcircuit resulted in the largest power draw of any subcircuit, with 49mW for the PMOS driver and 34mW for the NMOS driver. Due to lower hole mobility, PMOS sizing was adjusted to be relatively larger.

VIII. FET Sizing

The power fets are sizing accordingly to balance loss and well as supporting high current loads. The PMOS device was sized at 100um width and 300nm length, with a 20,000 multiplier resulting in a 0.6 mm² on-chip area. The NMOS was sized at half this due to electron mobility. At max load, the simulated losses were 36mW and 14mW, for PMOS and NMOS respectively.

IX. LC Sizing

We simulated using a 1μH inductor and 10μH capacitor. In a full system, these would be implemented off-chip, and the devices would have significant parasitics. For accuracy, we used real values of components. 1μH inductors rated for high current (such as Vishay Dale IHXL2000VZEB1R2M31) can have down to 15mΩ series resistance. Low voltage 0402 capacitors typically report 5mΩ of series resistance. We used these values in simulation to validate our circuit.

X. System Level Results

A. Steady-State Performance

The buck converter was stable in our load conditions, validated up to 3A. At this max load current, the device has a ripple of roughly 10mV with an average voltage output.

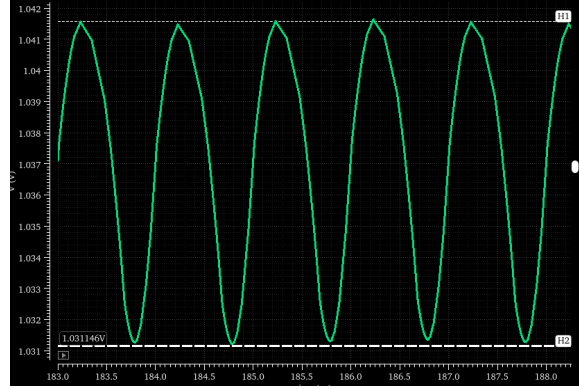


Figure 19. Transient response of buck converter.

B. Efficiency

The goal was to reach 90% efficiency across load conditions from 2A to 3A. This specification was achieved, and the power efficiency over load condition is shown in Figure 20.

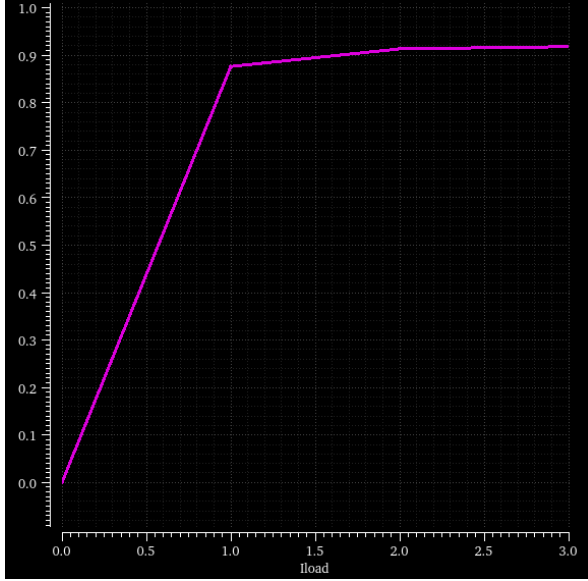


Figure 20. Power efficiency vs. load current, showing 90% efficiency at load currents greater than 1.7 A.

At low loads, the efficiency is dominated by quiescent current, and switching losses in the drivers whereas at high loads. At high loads, efficiency is dominated by losses in the parasitic resistance in the inductor, and losses in the PMOS and NMOS drive transistors, with some portion being contributed by the driver losses. The power loss distribution at 3A is shown below:

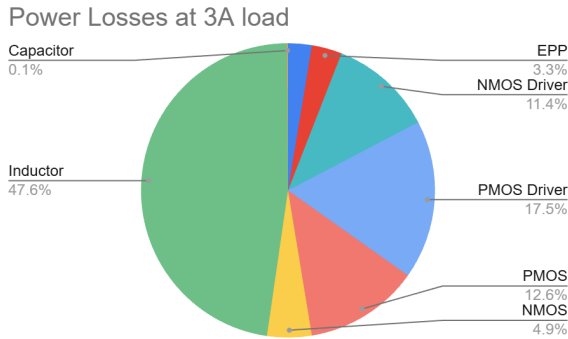


Figure 21. Causes of power loss.

Evidently, the main design tradeoff was losses in the PMOS/NMOS and losses in their respective drivers. To reduce PMOS/NMOS losses, large driver circuits were needed, but increased sizes also promoted switching losses. We sacrificed drive speed for efficiency, reducing the size and number of stages to promote lower power losses.

C. Load Step

To validate the performance of our circuit, a load step test was performed, stepping current from 2A to 3A. The circuit drooped 70mV during step, and raised 130mV on load drop, as shown below.

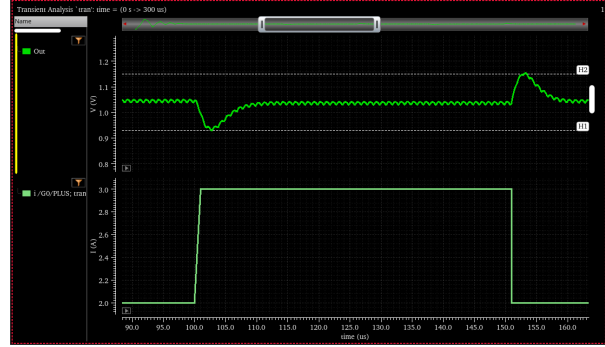


Figure 22. Load step response.

D. Line Step

Similarly to load step, line step was tested to ensure proper operation during power supply events. The output waveform, shown below, was stable, rising 0.1V on line step, and drooping 0.1V on line drop.

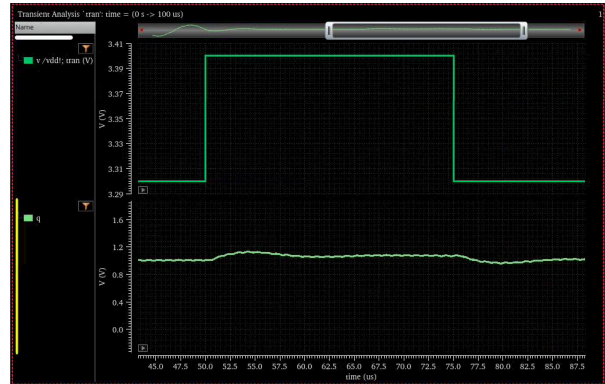


Figure 23. Line step response.

E. Power Supply Rejection Ratio

The PSRR was tested by injecting an AC stimulus at the input, and measuring the Periodic AC response on the output. The results were plotted across frequency, showing consistent -32dB PSRR across all load conditions. The results for $I_{load}=0.1$ A are shown in Figure 24:

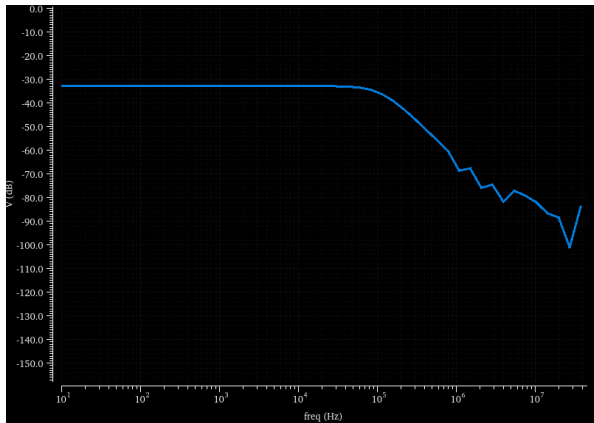


Figure 24. PSRR frequency response.

XI. Conclusion

For this project, we were successfully able to implement a 1MHz high efficiency, high load buck converter. We were >90% efficient in the load range of 2-3A. In addition, we had 1% (10 mV) ripple in steady state, our system was compensated using a DDA-based Type-III compensator to be stable, and said stability was proven during device response to load and line steps.

REFERENCES

- [1] L. Cheng, Y. Liu and W. -H. Ki, "A 10/30 MHz Fast Reference-Tracking Buck Converter With DDA-Based Type-III Compensator," in *IEEE Journal of Solid-State Circuits*, vol. 49, no. 12, pp. 2788-2799, Dec. 2014, doi: 10.1109/JSSC.2014.2346770.
- [2] N. M. Siu, P. K. T. Mok, K. Nang, Y.-H. Lam, and N. W.-H. Ki, "A voltage-mode PWM buck regulator with end-point prediction," *IEEE Transactions on Circuits and Systems II Analog and Digital Signal Processing*, vol. 53, no. 4, pp. 294–298, Apr. 2006, doi: <https://doi.org/10.1109/tcsii.2005.862024>.